

The Feasibility of Radio Frequency Attacks on Spacecraft and Ground Data Systems

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Abstract—A study was performed on the feasibility of a denial-of-service (DoS) attack on ground station networks for spacecraft telecommunications. Any spacecraft uplink and downlink are conducted through radio frequency (RF) and various modulation protocols, all of which do not have origin and destination identification characteristics. To simulate a spacecraft to ground station network, the tests were conducted using the ground data system from the F Prime open-source flight software and embedded systems framework developed by NASA’s Jet Propulsion Laboratory (JPL). Short range ultra-high frequency (UHF) radios operating at 915 MHz were used to simulate a DoS attack on a ground station. As a result, the communication protocol should be designed such that the software can discriminate between legitimate and malformed packets. Attempts to mitigate packet loss from RF noise was explored, including modifying preamble lengths, synchronization words, and hardware encryption.

I. FREQUENCIES ARE ON THE WEB!

Space exploration is an ever growing industry as more countries and organizations are populating our solar system with spacecraft. And with that, there is an increasing amount of RF traffic occurring across our space networks. To ensure there are no colliding signals, all spacecraft operating on RF have unique frequency bands regulated by the International Telecommunication Union (ITU) [1]. These frequencies are registered and publicly available on several databases such as [2] and [3]. However, this enables attackers to target a specific frequency and transmit packets towards the spacecraft or ground station, causing potential harm to hardware and/or software systems, and thus ending a mission before its scheduled termination. This ultimately leads to a loss of substantial funds, depending on the scale of the mission.

Current technologies exist to track RF signals [4]. However, these have not been tested in space. In the event that unwanted RF signals are detected in space, it

is difficult to trace the point of origin as the spacecraft is in continuous orbit.

II. RELATED WORK

Most spacecraft mitigate malicious communications, which includes harmful commands (i.e. shutdown or reboot commands) or packet replay attacks, by introducing encryption into their protocols. Various encryption algorithms are available that utilize the Advanced Encryption Standard (AES), but AES-CTR (Counter) and AES-GCM (Galois/Counter Mode) are among the few that are most reliable for radio cryptography. Both of these algorithms utilize private keys and randomized vectors called initialization vectors (IVs) to reduce the likelihood of cracking the private key through brute-force methods. The AES-GCM algorithm was implemented and tested during the development of NASA’s Hyperspectral Thermal Imager (HyTI) mission, which included X-Band, S-Band, and UHF space-rated radio modules [5].

The Space Attack Research and Tactic Analysis Project (SPARTA) [6] by the Aerospace Corporation analyzed potential threats to a spacecraft, including packet replays, geofencing, hardware/firmware corruption, flooding, or spoofing. Many of these attacks involve packet transmissions. Geofencing requires an attacker to acquire the location of the current spacecraft in orbit, which is often obtained from telemetry data. Hardware/firmware corruption is only possible when transmitting files to the spacecraft through the radio. Flooding requires an attacking spacecraft to generate RF noise by continuous transmissions of arbitrary packets. And spoofing requires attackers to modify packets to modify data and intended packet functionalities. Their recommendations against packet replays, geofencing, hardware/firmware corruption, and spoofing all include packet encryption and authentication. However, encryption was not a solution to flooding. Instead, their suggestion was to modify hardware to narrow the

range at which the radio is capable of accepting packets. However, this can also lead to the nullification of legitimate packets. Other suggestions involve increasing the security of the flight software.

Existing implementations to increase the security of flight software like encryption and software security may reduce the impact of denial-of-service (DoS) attacks, but does not prevent it entirely. The goal of this research was to mitigate DoS attacks at the hardware level, which is at a higher level compared to software security and encryption.

III. FRAMEWORK SECURITY AND STABILITY

Despite successful deployments of encryption into spacecraft telecommunication systems, attackers are still capable of transmitting unencrypted packets of randomized bytes towards the spacecraft and ground station. While these packets have meaningless information, they generate RF noise that will be received by the spacecraft and/or ground station, ultimately occupying the internal buffers of the radio(s) and increasing the decoding time, preventing legitimate packets from being processed.

RF attacks are to be expected, so a flight software framework should be designed to handle such attacks. One of the primary mitigation techniques within F-Prime involves a command dispatcher/handler, which ensures that all commands received are genuine. All commands are registered with an opcode and placed in a registration table upon initialization of the flight software [7]. If the command within a command packet is decoded but is not mapped to the registration table, it will be considered invalid and dropped. Likewise, if the flight software is unable to deserialize a packet to retrieve the opcode, it will be flagged as a malformed packet and dropped.

General packet frames, which encompasses commanding, telemetry, or file transfers, are commonly verified with a Cyclic Redundancy Check (CRC) checksum footer. In the event that a packet was modified, the checksum will fail and the packet will be dropped. The current F Prime flight software and ground data system (GDS) implementations utilize checksums to ensure packet communications are legitimate.

Additionally, all packet communications should follow a structured protocol for all trusted nodes in the spacecraft network. If a packet were to deviate from the expected protocol, it should be immediately dropped. The Consultative Committee for Space Data Systems (CCSDS) recommends standards for interfaces and protocols for space missions for communication effectiveness and compliance. A sample CCSDS recommended

standard packet header protocol can be found in [8]. Having a standardized data structure to create, store, and transfer variable-length data ensures packet identification and source/destination determination. Packet identifiers include, but not limited to, version numbers, type IDs, sequencing, and data length. Fig. 1 shows an example of a space packet header that follows the CCSDS recommendations.

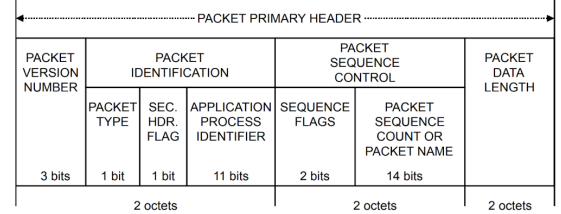


Fig. 1. Sample CCSDS recommended packet header.

Any packet analysis protocols, such as decoding, decrypting, or CRC checking, should minimize computation time to allow processing time to be allocated to other incoming packets. This ensures that if RF noise is present, the receiving station is capable of keeping up with the rate at which packets are being transmitted to it.

IV. ETHICAL CONCERNS

There are laws and regulations regarding RF use [9] set by the ITU which were followed throughout the procedures of this research. There are ranges of frequency bands, called ISM (Industrial, Scientific, and Medical) bands, intended for industrial, scientific, and medical use. The world is divided into regions where each region has its own allocation of frequency bands. The United States, including Alaska and Hawaii, is part of Region 2 and has been allocated with the 902 - 928 MHz frequency range. The UHF radio chosen for testing was the RFM69HCW set to operate at 915 MHz, which falls within the ISM frequency band and does not require an Amateur Radio License. Additionally, these radios are low-power and do not have long-range capabilities without a long-range antenna. The antennas chosen for the radios were low-range helical copper antennas with 2.15dBi of gain, a 50Ω impedance, and a maximum RF power output of 5 W.

V. EXPERIMENTAL PROCEDURE

To simulate a spacecraft and ground station network, three nodes were constructed utilizing RFM69HCW radios and Teensy 4.0 boards. All three nodes were identical: the radio was connected to the Teensy 4.0 through

the Serial Peripheral Interface (SPI). One node simulated a spacecraft, downlinking telemetry to the ground station, which was simulated using a second node. The ground station was also connected to a Linux/WSL host machine running JPL's F Prime GDS. The spacecraft will be running the F-Prime flight software framework and the ground station acted as a serial passthrough to the host machine. The third node simulated a malicious spacecraft, generating RF noise to attempt to block communications between the genuine spacecraft and the ground station. The rate at which the attacking spacecraft sent packets was gradually increased until the connection between the genuine spacecraft and ground was unstable or lost.

Since the spacecraft consistently sent telemetry down to the ground station, the number of packets being sent and received were constantly increasing. To accurately measure the number of packets sent by the spacecraft and received by the ground station, a command to send bursts of predefined "Hello World" packets was created. This command read two user-defined parameters: the number of packets to send and the interval, in milliseconds, at which those packets were to be sent. The telemetry of the F Prime GDS monitor shows the number of these predefined packets sent by the spacecraft and the number of those packets received by the ground station.

The procedures for the experiment were conducted to answer these questions:

1. How does the attacking spacecraft transmission rate affect packet loss?

During the transmission of the 1000 burst packets, the attacking spacecraft was configured to transmit packets of its maximum buffer size of 60 bytes at 0 Hz, 1 Hz, 10 Hz, 100 Hz, 200 Hz, 500 Hz, and 1000 Hz. This was repeated for each of the genuine spacecraft transmission rates described in Question 2 below. The distances between all three radios were negligible.

2. How does the genuine spacecraft transmission rate affect packet loss?

The rate at which the bursts of packets were sent could be configured as described earlier. The genuine spacecraft was configured to transmit at 1 Hz, 10 Hz, and 100 Hz.

3. How does distance of both the genuine and attacking spacecraft from the ground station affect packet loss?

To analyze how the distance of the attacking spacecraft from the ground station affects packet loss, the attacking

spacecraft was placed at various distances away: 0 m, 1.27 m, 2.54 m, 3.81 m, 5.74 m, 8.28 m, and 10.64 m. The genuine spacecraft had negligible distance between the ground station.

To analyze how the distance of both the genuine and attacking spacecraft from the ground station affects packet loss, the distances between both spacecraft were negligible and placed at various distances away: 0 m, 1.27 m, 2.54 m, 3.81 m, 5.74 m, 8.28 m, and 10.64 m.

In the worst case, both of these tests were conducted with a constant attacking rate of 1000 Hz.

These variables were kept constant throughout the experimental procedures:

1. Every burst consisted of 1000 packets.
2. All telemetry is being sent to the F Prime GDS monitor at 1 Hz.
3. The ground station radio listens to any available packets at a rate of 1000 Hz.
4. The frequency of all radios are set to 915 MHz.
5. The transmission power level of all radios are set to its rated maximum (+20 dBm).

The setup of the simulated spacecraft and ground station are shown in Fig. 2. The F Prime deployments for both the genuine spacecraft and ground station, as well as the standalone program for the attacking spacecraft can be viewed in [10].

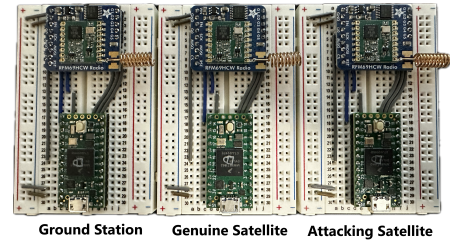


Fig. 2. Hardware setup for the experimental procedures.

VI. RESULTS

Telemetry values for the number of "Hello World" packets sent by the spacecraft and received by the ground station can be seen in the graphical user interface (GUI) of the F Prime GDS channels tab. A screenshot of these telemetry values are shown in Fig. 3.

During the experimental process, observations were made regarding the operations and behaviors of the RFM69HCW radio. The transmission of packets are prioritized over the reception of packets and thus, packet transmissions would always succeed despite the presence

Channel Id	Channel Name	Channel Value
0x10005	rfr69.NumHelloWorldSent	1000
0x10010004	gnd_rfr69.NumHelloWorldReceived	998

Fig. 3. F Prime GDS GUI sample telemetry

of RF noise. As a result, uplink commands would fail if RF noise was targeted at the genuine spacecraft. This was observed during the experiments: the genuine spacecraft could not be commanded to begin the transmission of "Hello World" packets. To bypass this, the command included a delay before the genuine spacecraft begins its transmission of "Hello World" packets to allow the attacking spacecraft to initiate its DoS attacks.

The results for the percentage of packet lost after sending bursts of 1000 packets at various attack rates from the malicious spacecraft and various transmission rates from the genuine spacecraft is shown in Fig. 4. It can be seen that as the rate at which the attacking spacecraft sends noise packets increases, the percentage of packets lost at the ground station increases as predicted. Additionally, the rate at which packets are downlinked to the ground station does not affect the probability of packet loss: transmission rates of 1 Hz, 10 Hz, and 100Hz all yield similar packet loss percentages.

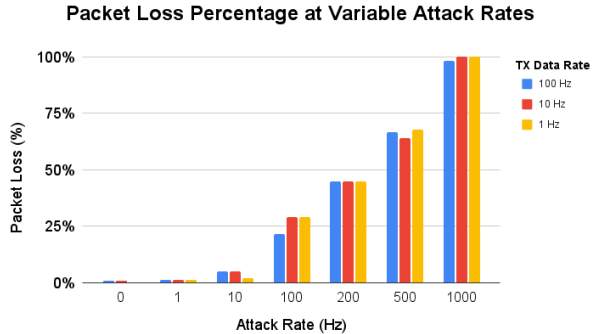


Fig. 4. Packet loss percentage at variable attack and transmission rates.

The results for the percentage of "Hello World" packets lost at variable attacking distances when the genuine spacecraft remained at a negligible distance from the ground station and transmitted at a rate of 1 Hz, 10 Hz, and 100 Hz are shown in Fig. 5. While the percentage of packet losses were not consistent throughout the trials, the general trend shows that as the attacking distance increases, the percentage of packet lost decreases.

This was to be expected as the RF signals of the genuine packets were received before the signals that originated from the attacking spacecraft.

Packet Loss Percentage at Variable Attacking Distances

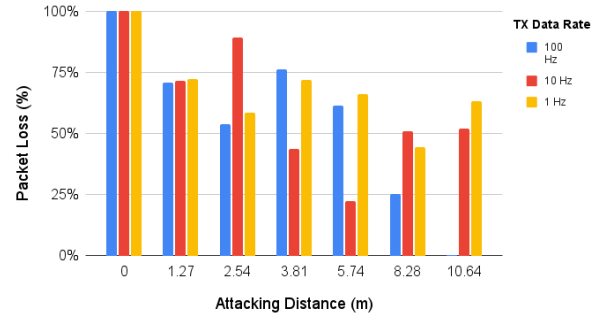


Fig. 5. Packet loss percentage at variable attacking distances.

The results for the percentage of "Hello World" packets lost at variable distances of both the genuine spacecraft and attacking spacecraft from the ground station are shown in Fig. 6. Similar to Fig. 5, the percentage of packet lost were not consistent throughout the trials, but the general trend showed that as long as the attacking spacecraft remains a similar distance from the ground station as the genuine spacecraft, the percentage of packet lost remains high.

Packet Loss Percentage at Variable Spacecraft Distances

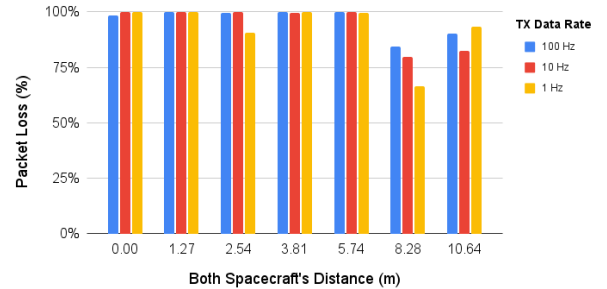


Fig. 6. Packet loss percentage at variable distances of both the genuine and attacking spacecraft.

VII. DOS MITIGATION

Various denial-of-service (DoS) mitigation methods are available, three of which could be tested using the RFM69HCW radios: variable preamble lengths, different synchronization words, and hardware encryption.

A. Preamble

Preamble lengths of a radio packet are one of the few crucial properties of packet transmissions as they are

used for synchronization. A preamble can be thought of as a series of bytes at the beginning of every packet that never change. These bytes are a series of alternating 1s and 0s. For example, a preamble length of 4 would be "1010". When a radio recognizes these exact series of bytes in its buffer, it indicates that a packet was received. Upon receiving another instance of those exact series of bytes, that indicates that a second packet was received. The radio can then slice all the bytes between those two preamble bytes, which becomes the raw data that was sent, and return them to the user. The radios communicating must have identical preambles to correctly indicate the start of a packet.

To test whether different preamble lengths affect the packet loss rate, the genuine spacecraft and ground station were set to have a preamble of 16 bytes while the attacking spacecraft has a preamble of 4 bytes. In theory, the ground station would be looking for the exact series of those 16 bytes to be considered a valid packet and drops every other packet. In the worst case, the attacking spacecraft was configured to transmit at a rate of 1000 Hz and the genuine spacecraft was configured to transmit at 1 Hz, 10 Hz, and 100 Hz. The results of this test are shown in Fig. 7.

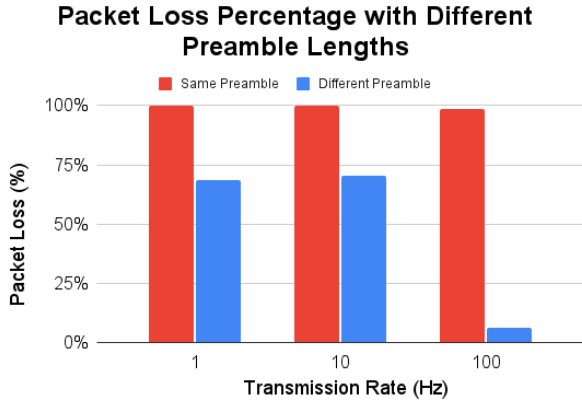


Fig. 7. Packet loss percentage when the attacking spacecraft transmits at a rate of 1 kHz using the same (blue) and different (red) preamble lengths. The tests were done at various transmission rates from the genuine spacecraft.

The red bars in Fig. 7 was obtained from the same experimental results as Fig. 4, which was when the attacking spacecraft transmitted at 1000 Hz with the same preambles as the other two radios. The packet loss was at 100% for this case. However, the blue bars shows that having different preamble lengths for the same transmission rates decreases the number of packets lost. When transmitting at 100 Hz, the packet loss was found

to be only 6.30% which was significantly lower than the 1 Hz and 10 Hz transmission rates. This is most likely because having a higher transmission rate increases the likelihood of multiple simultaneous preambles to be the 16 bytes. A lower transmission rate would result in a mixture of 16-byte and 4-byte preambles. For example, if three 16-byte preambles were received in order, that would result in two valid packets. However, if the first preamble received was 16 bytes, then the next was 4 bytes, and the next was 16 bytes, no packets would be considered valid.

B. Synchronization Words

Synchronization words are another crucial property of packet transmissions. They behave similar to the preamble as they are used for packet synchronization. The synchronization word is a series of bytes placed in the beginning of a packet to mark the start of a packet. The receiving radio checks for these specific bytes before storing the raw bytes into its buffer. This differs from the preamble because packets that do not contain those bytes are completely ignored by the radio. Therefore, packet losses are expected to be lower compared to the previous test regarding preamble lengths. Radios generally have a low number of bytes allocated for their synchronization word. The RFM69HCW used throughout these experiments has a range from 0-4 bytes.

To test whether different synchronization words affect the packet loss rate, the genuine spacecraft and ground station were configured to have a synchronization word of [0x12, 0x34, 0x56, 0x78] while the attacking spacecraft was configured to be [0x2D, 0xD4]. In the worst case, the attacking spacecraft was configured to transmit at a rate of 1000 Hz and the genuine spacecraft was configured to transmit at 1 Hz, 10 Hz, and 100 Hz. The packet loss when all spacecraft have identical synchronization words were nearly 100%. However, having different synchronization words resulted in packet losses of 0.1%, 0%, and 0% when the genuine spacecraft transmitted at 1, 10, and 100 Hz, respectively. This showed that having different synchronization words completely mitigates DoS attacks.

C. Hardware Encryption

Some radios incorporate hardware encryption which can be configured through software. Integrated encryption engines include checksum validations (i.e. CRC-16 or CRC-32), where if the checksum fails after decryption then the packets are dropped. This differs from software level encryption algorithms because the point where

packets are dropped occur earlier for hardware encryption than software encryption. For software encryption, all packets, whether encrypted or not, will always be accepted and then decrypted which takes significantly more computation time.

To test the reliability of hardware-level encryption to mitigate RF noise, the genuine spacecraft and ground station were configured to have identical encryption keys while the malicious spacecraft will continue to transmit unencrypted packets. For this test, the malicious spacecraft transmitted at 1000 Hz while the genuine spacecraft transmission rate was configured to be 1 Hz, 10 Hz, and 100 Hz. The distances between all three radios were negligible.

The results of this test is shown in Fig. 8. The red bars were obtained from the same experimental results as Fig. 4 where all radios did not incorporate encryption. The blue bar is the result from the encrypted packets. Despite incorporating hardware encryption, the attacking spacecraft successfully caused a DoS to the ground station with relatively high packet loss. It was observed that the ground station radio was still receiving packets from the attacking spacecraft into its buffer despite having no encryption key. While the specifications of the RFM69HCW indicated an integrated AES-128 and CRC-16 packet engine, this did not appear to behave as expected. The checksum should not have passed as the decryption should have generated a malformed packet.

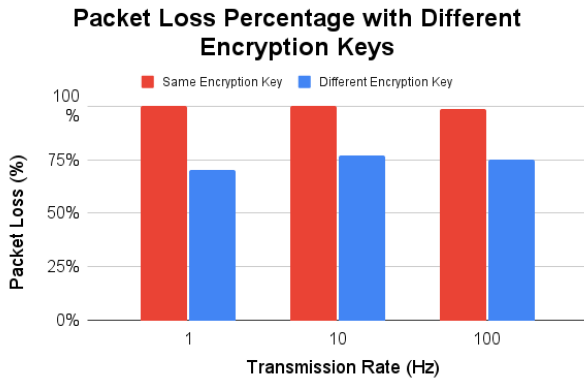


Fig. 8. Packet loss percentage when the attacking spacecraft transmits at a rate of 1 kHz using the same (blue) and different (red) encryption private keys. The tests were conducted at various transmission rates from the genuine spacecraft.

VIII. EXPERIMENT LIMITATIONS

The hardware used throughout these trials were low cost UHF radios with a data rate of 300 kb/s. The results obtained do not necessarily apply to all radios.

Some radios are capable of transmitting as high as 3 Mb/s which would overpower an attacking spacecraft transmitting at a lower data rate.

IX. CONCLUSION

Denial-of-service attacks on spacecraft and/or ground stations, while unethical and illegal, can be a common attack as it is difficult to trace in space. Every registered space mission is publicly available and exposes the frequency bands being used. Because of this, the flight and ground software design is crucial to mitigate such attacks. The experiments conducted in this paper proved that as long as a malicious user gains knowledge of a spacecraft's frequency band, location of a spacecraft in orbit, and/or location of ground stations operating at said frequencies, they are able to introduce RF noise in the vicinity and increase the probability of packet loss. Only unique radio configurations, like preamble length, synchronization bytes, or hardware encryption, can be classified as proprietary to mitigate packet loss due to RF noise as users will not have access to such information.

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